METEHAN ALP MEMIS

Data-Centric AI · Interpretable Machine Learning · Transportation Geotechnics · Civil Infrastructure

Urbana-Champaign, Illinois, USA | +1 314-637-8290 | metehanalp.memis1@outlook.com | linkedin.com/in/metehanalpmemis | Google Scholar

RESEARCH PROFILE

Ph.D. researcher at the University of Illinois Urbana-Champaign working at the intersection of data-centric and interpretable machine learning, transportation geotechnics, and civil infrastructure. Research emphasizes reliable learning from large, heterogeneous engineering datasets, leakage-resistant validation, geographic and environmental generalization, and physically meaningful interpretation. Recent work used 82,682 laboratory measurements from 603 LTPP pavement sections across 40 U.S. states to develop and evaluate interpretable machine-learning models for resilient modulus prediction.

RESEARCH INTERESTS

Data-Centric AI · Trustworthy & Explainable ML · Large-Scale Infrastructure Data · Dataset Quality & Generalization · Transportation Geotechnics · Pavement Materials & Performance · Climate-Resilient Infrastructure · Seismic & Critical-Infrastructure Resilience

EDUCATION

Ph.D., Civil & Environmental Engineering - Transportation Geotechnics Aug 2026-Present

• University of Illinois Urbana-Champaign (UIUC), Urbana, IL | Advisor: Prof. Erol Tutumluer | Alfred Korn Fellowship

M.S., Civil Engineering - Geotechnical Engineering Aug 2024-Present

• Southern Illinois University Edwardsville (SIUE), Edwardsville, IL | GPA: 3.86/4.00 | Computational geotechnics, THM soil behavior, dynamic soil characterization

- Degree requirements completed; formal conferral expected December 2026

M.S., Civil Engineering - Geotechnical Engineering Sep 2021-Feb 2024

• Karabuk University, Turkiye | GPA: 3.64/4.00 | Thesis: Prediction of Soil Strength and Dynamic Properties Using Dynamic Cone Penetration (DCP) Data | TUBITAK/KBUBAP-funded

• Erasmus+ Exchange, ESTP Paris, France (2022-2023): Civil & Nuclear Structures, Bridge Design, Geotechnical & Structural Engineering

B.S., Civil Engineering Sep 2016-Jul 2021

• Karabuk University, Turkiye

SELECTED RESEARCH HIGHLIGHTS

• Large-scale, leakage-resistant infrastructure ML: developed an interpretable resilient-modulus prediction framework using 82,682 measurements from 603 independent LTPP pavement sections across 40 states; used section-based train/test partitioning to prevent leakage and SHAP to assess physically meaningful drivers.

• Climate and domain heterogeneity: evaluated model behavior across climate zones and heterogeneous infrastructure conditions, highlighting the importance of data composition and realistic out-of-section generalization.

• AI for climate-resilient infrastructure: published work on machine-learning approaches to permafrost degradation and infrastructure implications, integrating geotechnical domain knowledge with data-driven modeling.

• Data-driven critical-infrastructure assessment: conducted global seismic site characterization of nuclear power plant sites using USGS Vs30 data and developed research on AI-supported post-disaster ground-damage assessment.

HONORS, AWARDS & RESEARCH FUNDING

Alfred Korn Fellowship - University of Illinois Urbana-Champaign, CEE 2026

Research Grant for Graduate Students (RGGS) Award - SIUE Graduate School 2026

• Project: Nonlinear Seismic Site Response Analysis of CPT-Based Soil Profiles: A Comparative Study of Near-Field and Free-Field Conditions in Iskenderun.

MPOWER Monthly Scholarship - MPOWER Financing 2026

TUBITAK / KBUBAP Research Funding (KBUBAP-23-YL-089) 2023-2024

PEER-REVIEWED JOURNAL ARTICLES

1. **Memis, M.A.**, & Ulus Memis, S. (2026). Interpretable machine learning framework for resilient modulus estimation of unbound granular base layers: Insights from the LTPP database. Transportation Geotechnics, 64, 102292. https://doi.org/10.1016/j.trgeo.2026.102292
2. **Memis, M.A.**, & Ulus Memis, S. (2026). Global seismic site characterization of nuclear power plant sites using USGS Vs30 data. Nuclear Engineering and Technology, 104595. https://doi.org/10.1016/j.net.2026.104595
3. **Memis, M.A.**, & Ulus Memis, S. (2026). Resilience framework for critical infrastructure: Artificial intelligence applications in post-disaster ground damage assessment. Journal of Engineering in Industrial Research, 7(4), 195-211.
4. **Memis, M.A.**, & Ulus Memis, S. (2026). Monitoring land surface temperature (LST) changes in Northern Alaska's permafrost regions using MODIS data. Journal of Engineering in Industrial Research, 7(3), 91-99. https://doi.org/10.48309/jeires.2026.561167.1325
5. **Memis, M.A.**, & Ulus Memis, S. (2026). Bridging artificial intelligence and geotechnical engineering through education: A social framework for disaster risk awareness and preparedness. Canadian Journal of Educational and Social Studies, 6(1), 114-125. https://doi.org/10.53103/cjess.v6i1.443
6. **Memis, M.A.**, Keskin, I., Demir, S., & Ulus Memis, S. (2025). Machine learning-based prediction of permafrost degradation and its implications on geotechnical infrastructure: A comprehensive review. AI in Civil Engineering, 4(1), 28. https://doi.org/10.1007/s43503-025-00080-8
7. Keskin, I., & **Memis, M.A.** (2025). Prediction of soil strength and dynamic properties through the dynamic cone penetration index. Discover Civil Engineering, 2(1), 142. https://doi.org/10.1007/s44290-025-00305-w
8. Keskin, I., & **Memis, M.A.** (2025). Stability analyses of rock slopes and improvement solutions: A case of Mugla Bakicak, Turkiye. Journal of Engineering Research and Reports, 27(7), 1-27. https://doi.org/10.9734/jerr/2025/v27i71560

MANUSCRIPTS UNDER REVIEW / IN REVISION

1. **Memis, M.A.**, et al. A compound seismic screening index for global nuclear power plant sites combining probabilistic rock-PGA hazard with Vs30-based site conditions. Discover Hazards. Under revision.
2. **Memis, M.A.**, et al. Machine learning based prediction of pavement deterioration risk under climate variability: A national-scale analysis using the LTPP database. International Journal of Pavement Research and Technology. Major revision.
3. **Memis, M.A.**, et al. Trustworthy explainable artificial intelligence in geotechnical engineering: Methods, validation practices, research gaps, and a proposed reporting framework. Civil Engineering Science and Technology Journal. Revision with editor.
4. **Memis, M.A.**, et al. Enhancing critical infrastructure resilience in seismic-volcanic regions through an integrated geotechnical modeling framework: A transferable methodology from a Turkiye case study. Bulletin of Engineering Geology and the Environment. Under review.
5. **Memis, M.A.**, et al. Advancements in rockfall risk management: A systematic review of identification, modeling, and mitigation strategies. Discover Civil Engineering. Under revision.
6. **Memis, M.A.**, et al. Predicting permafrost degradation and infrastructure risk in cold regions using machine learning approaches. Discover Geoscience. Under revision.

RESEARCH EXPERIENCE

Graduate Research Assistant - Geotechnical Engineering, SIUE Aug 2025-Aug 2026

- Analyzed climate-sensitive geosystems and thermo-hydro-mechanical soil behavior using Python (pandas, NumPy) and PLAXIS; integrated computational analysis with peer-reviewed research development.
- Conducted nonlinear seismic site-response research using CPT-based profiles and engineering data, including comparison of near-field and free-field conditions.

Graduate Research & Teaching Assistant - Saint Louis University Jan 2025-Aug 2025

- Processed and analyzed experimental geotechnical data in Python; conducted triaxial, direct shear, consolidation, and bender-element testing for static and cyclic soil characterization.
- Performed geotechnical site investigation and USCS soil classification per ASTM D2487; supported undergraduate geotechnical laboratory instruction.

Research Assistant - Geotechnical Engineering, Karabuk University May 2023-Feb 2024

- Built regression-based predictive models linking in-situ DCP measurements to CBR, unconfined compressive strength, cohesion, friction angle, Gmax, and Vs.
- Implemented finite-element simulations in PLAXIS and ABAQUS for soil-structure interaction within TUBITAK/KBUBAP-funded research.

PROFESSIONAL & LEADERSHIP EXPERIENCE

Transportation Engineer - Erbil Engineering / General Directorate of Highways (KGM) May 2024-Jan 2025

- Designed roadway alignments, cross-sections, and earthwork using Civil 3D and MicroStation InRoads; performed geotechnical site investigations and slope-stability analyses for transportation infrastructure.

Project Engineer - Ronesans Holding / Yazici Architectural & Engineering Jul 2021-Jul 2022

- Supported geotechnical/structural design and coordinated civil engineering activities on large-scale construction projects.

Co-Founder - Ankademia, AI-powered academic matching platform 2025-Present

- Co-founded a platform applying AI to connect students and researchers with academic opportunities and mentorship.

CONFERENCE PROCEEDINGS

1. Keskin, I., & **Memis, M.A.** (2024). Estimation of strength and plasticity properties of soils with DCP data. 7th International Azerbaijan Congress on Life, Engineering & Mathematical Sciences.
2. Keskin, I., & **Memis, M.A.** (2024). Estimation of shear and dynamic properties of soils with DCP data. 8th International "Artemis" Scientific Research Congress.
3. Kahraman, S., **Memis, M.A.**, & Keskin, I. (2024). Kaya sevlerinin stabilite analizleri ve iyilestirme cozumleri: Mugla/Golturkbuku ornegi. 2nd International Palestra Scientific Research Congress.
4. Apul, S., Keskin, I., **Memis, M.A.**, & Ekemen Keskin, T. (2024). An evaluation on the effect of water chemistry on soil hydraulic properties. 18th International Istanbul Congress on Life, Engineering & Architecture.
5. Necim, A., Keskin, I., & **Memis, M.A.** (2023). Effect of thermal effect on Atterberg limits: A case of bentonite-sand mixed soil. International Asian Economy and Scientific Research Congress, 163-170.

OTHER SCHOLARLY OUTPUT

1. **Memis, M.A.**, & Keskin, I. (2026). Dynamic cone penetration testing (DCP) in geotechnical design: Applications, limitations, and innovations for infrastructure assessment. Zenodo (technical report/preprint).
2. **Memis, M.A.** (2024). Estimation of some engineering properties of soils by dynamic cone penetrometer (DCP) [M.S. thesis, Karabuk University].

TECHNICAL SKILLS

Machine Learning & Data Science: Python (pandas, NumPy, Matplotlib, SciPy, scikit-learn), SHAP / interpretable ML, statistical modeling, MATLAB, Tableau, advanced Excel
Numerical & Geotechnical Modeling: PLAXIS 2D/3D, ABAQUS, DEEPSOIL, GeoStudio, Phase2, RocFall 2D/3D, Settle3, SAP2000
Remote Sensing & GIS: Google Earth Engine, MODIS, QGIS, UAV photogrammetry, Structure-from-Motion, DEM generation, Dips 7.0
Civil Engineering & Testing: CPT/DCP interpretation, seismic site response, slope stability, foundation design; triaxial, direct shear, consolidation, compaction, bender element; AutoCAD, Civil 3D, MicroStation InRoads, Revit